

Delivery Performance, Delay Risk, and Logistics Efficiency Analysis in Global Supply Chain Operations

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Abstract

This study analyzes delivery performance and logistics efficiency using 180,519 shipment/order records. A Delay Gap was calculated as actual shipping days minus scheduled shipping days. Shipments were classified as Delayed, On-time, or Early, while the dataset's Late_delivery_risk field was retained as a separate operational risk measure. The analysis found that 57.28% of records had a positive Delay Gap, the average Delay Gap was approximately +0.57 days, and 54.83% were classified as late-delivery risk. First Class and Second Class showed substantially higher delay rates than Standard Class, while delay rates across major markets remained close to 57%. Customer segments showed only small differences. An interactive Streamlit dashboard was developed to communicate these findings through KPI cards, filters, and comparative analyses.

1. Introduction

Reliable delivery is a core requirement of global supply-chain operations. Delays can affect customer satisfaction, inventory planning, transportation utilization, and profitability. Data analytics provides a practical way to measure delivery performance, compare service modes, identify geographic patterns, and support continuous improvement. This project evaluates delivery performance in a global supply-chain dataset associated with APL Logistics. The study focuses on delivery timing, late-delivery risk, shipping-mode efficiency, market and regional differences, customer segments, and selected financial indicators. The final results are presented through an interactive Streamlit dashboard.

2. Objectives

The objectives are to clean and validate shipment data; calculate Delay Gap; classify shipments as Delayed, On-time, or Early; measure overall delivery KPIs; compare shipping modes, markets, regions, and customer segments; examine the relationship between delivery risk and financial measures; and develop an interactive dashboard for operational decision support.

3. Dataset and Data Preparation

The dataset contains 180,519 records and 40 columns. Important fields include Days for shipping (real), Days for shipment (scheduled), Shipping Mode, Market, Order Region, Customer Segment, Delivery Status, Late_delivery_risk, Sales, and Order Profit Per Order. Validation found no duplicate rows. Only 11 missing values were identified: 8 in Customer Lname and 3 in Customer Zipcode. These fields were not required for the core delivery calculations. The main derived metric is: $\text{Delay Gap} = \text{Days for shipping (real)} - \text{Days for shipment (scheduled)}$. A positive gap indicates a delay, zero indicates on-time performance, and a negative gap indicates early shipment. Delivery classes were defined as Delayed (>0), On-time ($=0$), and Early (<0). Late_delivery_risk was analyzed separately because it does not exactly match the Delay Gap definition.

4. Methodology

The analysis followed a descriptive-analytics workflow: data loading and validation, metric creation, delivery classification, KPI calculation, grouped analysis, financial comparison, and dashboard development. Principal KPIs were: On-Time/Early Rate = percentage with Delay Gap ≤ 0 ; Delay Rate = percentage with Delay Gap > 0 ; Average Delay = mean Delay Gap; Late Delivery Risk = mean Late_delivery_risk $\times 100$. Grouped analyses used order count, average delay, delay rate, and late-delivery risk for Shipping Mode, Market, Order Region, and Customer Segment. Correlation was used only as a descriptive measure and does not establish causation.

5. Results and Findings

Overall, 103,400 records (57.28%) had a positive Delay Gap, 33,753 (18.70%) were on time, and 43,366 (24.02%) were early. The average Delay Gap was approximately +0.57 days and the median was +1 day. Late_delivery_risk identified 98,977 records (54.83%) as late-delivery risk. Shipping mode produced the strongest contrast. First Class had 27,814 orders, a 100% positive Delay Gap rate, an average gap of about +1.00 day, and late-delivery risk of about 95.32%. Second Class had 35,216 orders, an average gap of about +1.99 days, a delay rate of about 79.73%, and risk of about 76.63%. Same Day had 9,737 orders, an average gap of about +0.48 days and risk of about 45.74%. Standard Class had the largest volume, 107,752 orders, an average gap close to zero, a delay rate of about 39.77%, and risk around 38.07%. At market level, delay rates were consistently high: Europe 57.69%, Pacific Asia 57.32%, USCA 57.15%, LATAM 57.02%, and Africa 56.81%. Among regions with substantial volume, Central Africa had a delay rate of about 60.70%, followed by Western Europe at 58.52%, South Asia at 58.51%, South of USA at 58.10%, Southeast Asia at 57.98%, and East of USA at 57.98%. Customer segments were very similar: Home Office 57.52%, Consumer 57.30%, and Corporate 57.11% delay rate. Financially, total sales were approximately 36.78 million and total order profit approximately 3.97 million. About 18.71% of records had negative profit. Sales and profit had almost no linear relationship with late-delivery risk, with correlations close to -0.004.

6. Discussion

The findings indicate that delivery performance is a significant operational issue in the analyzed data. More than half of records exceeded scheduled shipping duration. The consistency across markets suggests that broad process factors may be more important than a single geographic bottleneck. The poor performance of First Class and Second Class requires investigation. Possible explanations include tight service commitments, capacity constraints, routing, fulfillment processes, or data-definition issues. These possibilities cannot be confirmed from the dataset alone and should be tested with carrier, route, warehouse, and timestamp data. The weak relationship between financial value and late-delivery risk also suggests that delivery problems are distributed across orders rather than limited to high-value transactions.

7. Business Recommendations

1. Investigate First Class and Second Class at carrier, route, warehouse, and fulfillment-center level.
2. Review whether scheduled shipping durations are realistic for each service mode.
3. Prioritize high-volume regions as well as regions with high delay rates.
4. Keep Delay Gap and Late_delivery_risk as separate KPIs with clear definitions.
5. Add carrier, route, warehouse, pickup timestamp, and delivery timestamp data for root-cause analysis.
6. Collect reliable date/time fields to support future weekly and monthly trend analysis.
7. Establish a recurring KPI dashboard for delay rate, average delay, on-time/early rate, and late-delivery risk.

8. Dashboard Implementation

An interactive Streamlit dashboard was developed using Python and pandas. It provides KPI cards, shipping-mode comparison, market analysis, regional analysis, customer-segment analysis, delivery-status analysis, and Delay Gap distribution. Sidebar filters allow users to select Shipping Mode, Market, Order Region, and Customer Segment, and the results update according to the selected filters. The dataset does not contain a reliable genuine order/delivery date field suitable for a date-range filter. Therefore, no artificial date filter was added.

9. Limitations

The study is descriptive and does not establish causal relationships. The dataset lacks several variables needed for root-cause analysis, including carrier, route, warehouse, weather, customs events, and reliable timestamps. Delay Gap and Late_delivery_risk are not identical measures and must be interpreted separately. The findings should therefore be validated against current operational records before real-world service-level decisions are made.

10. Conclusion

This study demonstrates how shipment data can be transformed into practical logistics performance indicators. The main findings are a 57.28% positive Delay Gap rate, an average Delay Gap of approximately 0.57 days, and a 54.83% late-delivery risk rate. First Class and Second Class showed the highest operational risk, while delay rates were consistently high across markets and customer segments showed little variation. The Streamlit dashboard converts these findings into an interactive decision-support tool. Future work should incorporate reliable timestamps, carrier and route data, warehouse information, and predictive analytics to identify the causes and probability of future delays.

References

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